

TROITSK, Russian Federation

WMO: 287480

Lat: 54.083N

Lon: 61.617E

Elev: 630

StdP: 14.36

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.8	-18.1	-29.3	1.1	-23.5	-23.7	1.5	-17.9	23.1	21.4	22.0	21.1	2.2	270	0.596

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	89.4	66.0	86.0	65.2	83.0	64.1	69.2	83.2	67.7	81.2	66.2	78.8	8.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.6	93.4	74.5	63.1	88.4	72.7	61.6	83.7	71.8	33.7	83.4	32.5	81.1	31.2	78.8	78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.5	18.8	15.9	-30.6	94.4	6.2	3.6	-35.1	97.0	-38.7	99.2	-42.1	101.2	-46.6	103.8		
(5)			-30.9	71.7	6.1	2.6	-35.2	73.6	-38.8	75.1	-42.2	76.5	-46.7	78.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.2	4.5	7.2	21.4	42.0	56.3	65.4	67.6	64.9	54.2	40.2	23.0	10.1
	DBStd	25.03	11.97	12.24	9.92	10.19	8.92	8.06	7.03	7.70	8.20	8.80	12.52	12.72
	HDD50	6236	1411	1198	888	281	41	4	0	1	44	320	811	1237
	HDD65	10145	1876	1618	1353	692	293	92	55	99	336	768	1261	1702
	CDD50	1941	0	0	0	39	237	467	546	463	172	17	0	0
	CDD65	373	0	0	0	0	23	104	137	96	13	0	0	0
	CDH74	4124	0	0	0	18	384	1173	1404	976	169	0	0	0
	CDH80	1389	0	0	0	2	90	415	473	355	54	0	0	0
	WSAvg	6.4	5.4	6.0	7.1	7.6	7.5	6.8	5.7	5.6	6.0	6.7	6.4	5.9
	PrecAvg	14.9	0.7	0.7	0.8	0.9	1.5	1.9	2.4	1.9	1.0	1.2	0.9	0.9
(15) Precipitation	PrecMax	19.2	1.5	1.9	1.9	2.4	3.5	4.2	5.1	6.5	2.2	3.8	2.0	2.2
	PrecMin	8.7	0.0	0.1	0.1	0.0	0.0	0.5	0.6	0.4	0.1	0.1	0.1	0.2
	PrecStd	3.2	0.4	0.4	0.6	0.6	0.9	1.1	1.2	1.3	0.6	0.8	0.6	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.2	35.9	48.1	76.7	86.4	92.5	94.3	92.6	87.1	69.2	50.2	36.9
		MCWB	32.2	33.3	41.5	55.5	60.4	65.8	67.7	67.4	60.5	51.3	44.7	33.4
	2%	DB	29.7	32.7	40.9	69.5	82.6	88.6	88.4	88.2	78.3	63.6	44.5	33.8
		MCWB	28.2	30.2	35.8	52.4	59.7	64.8	66.7	66.5	58.2	50.2	40.4	31.4
	5%	DB	26.2	29.1	37.3	64.3	78.6	84.9	85.0	84.0	73.0	58.8	41.0	30.9
		MCWB	24.6	27.3	33.9	50.1	58.1	63.8	66.7	65.2	57.6	47.6	37.4	28.9
	10%	DB	23.3	25.5	34.9	59.0	73.8	81.0	81.9	79.4	68.3	53.9	37.9	27.7
		MCWB	22.0	23.7	32.3	47.4	55.8	63.0	65.8	63.8	55.7	45.7	35.1	26.0
	0.4%	WB	32.4	33.8	42.6	56.6	64.5	69.9	71.5	70.7	63.8	54.9	45.0	34.1
		MCDB	34.0	36.0	48.1	72.0	78.9	86.3	84.0	85.7	81.5	64.4	47.8	36.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	28.3	30.3	36.4	53.7	61.7	67.4	69.8	68.4	60.5	51.6	41.0	31.7
		MCDB	29.9	32.5	40.0	66.7	77.5	82.0	83.1	83.4	73.2	61.4	44.2	33.6
	5%	WB	24.7	27.4	34.3	50.9	59.6	65.7	68.4	66.8	58.4	48.7	37.9	29.0
		MCDB	26.1	29.1	36.7	62.7	74.7	79.8	81.2	80.5	70.0	57.0	40.5	30.9
	10%	WB	22.0	23.7	32.6	48.1	57.4	64.1	66.9	65.0	56.4	46.2	35.2	26.0
		MCDB	23.4	25.5	34.9	58.4	71.3	78.0	78.9	76.3	67.0	53.1	37.7	27.7
	5% DB	MDBR	14.6	17.1	16.3	18.3	21.5	20.9	19.0	18.9	19.0	15.9	12.8	13.7
		MCDBR	14.5	14.4	15.3	27.2	28.9	26.3	24.4	25.7	26.3	25.1	14.2	12.7
		MCWBR	13.6	13.1	11.7	13.4	11.7	9.5	8.5	9.0	11.5	13.7	10.7	11.7
		MCWBR	14.1	14.2	13.9	24.9	25.5	23.0	21.3	22.5	22.7	22.2	12.9	12.6
(40) Clear-Sky Solar Irradiance	taub		0.285	0.316	0.370	0.412	0.409	0.406	0.427	0.444	0.386	0.355	0.315	0.285
		taud	2.080	2.040	1.972	2.119	2.228	2.312	2.242	2.169	2.285	2.266	2.192	2.067
	Ebn at Noon		202	236	248	257	266	268	259	245	245	223	191	173
		Edh at Noon	22	32	44	43	41	38	41	41	32	26	20	18
(44) All-Sky Solar Radiation	RadAvg		323	674	1162	1448	1729	1888	1801	1447	1015	542	280	207
	RadStd		24	33	84	149	162	156	124	142	79	85	29	31

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page